

AmirArsalan Rezaianzadeh

Python Developer | Machine Learning | LLM & Generative AI | Computer Vision

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SUMMARY

Computer Science student (6th semester) at Islamic Azad University, Science and Research Branch, with hands-on experience in Python, machine learning, and computer vision. Built real-world projects including a YOLOv8-based security system with face recognition and an ML pipeline for customer churn prediction. Currently expanding into Large Language Models (LLMs) and Generative AI — building a production-oriented Retrieval-Augmented Generation (RAG) chatbot as part of a self-directed, structured learning path covering transformer architectures, Hugging Face, LangChain, and backend integration with Django.

SKILLS

Programming: Python (Advanced), SQL

AI & Machine Learning: Machine Learning (Scikit-learn), Computer Vision (OpenCV, YOLO), Face Recognition (dlib), Data Analysis (Pandas, NumPy)

LLM & Generative AI (In Progress): Transformer Architecture & Attention Mechanisms, PyTorch, Hugging Face Transformers, LangChain, RAG Pipelines (Embeddings, Vector Databases), Prompt Engineering, LoRA/QLoRA Fine-tuning

Web & Backend: Django, Django REST Framework, Flask, FastAPI (In Progress)

Tools & Other: Git & GitHub, Docker, PostgreSQL, Tkinter, Jupyter Notebook, LabelImg

PROJECTS

AI-Powered Business Chatbot (RAG-based) — In Progress | Python • PyTorch • Hugging Face • LangChain • Django • Vector DB

- Building an end-to-end conversational AI assistant for small businesses, combining a Django/DRF backend with a Retrieval-Augmented Generation (RAG) pipeline for domain-specific question answering.
- Implementing document ingestion and chunking, embedding generation, and vector similarity search (Chroma / pgvector) to ground LLM responses in business-specific knowledge and reduce hallucination.
- Studied and implemented transformer internals from scratch (tokenization, self-attention, multi-head attention, training loop) to build a strong foundation before working with pretrained models.
- Designing the system to support both hosted LLM APIs and locally-hosted open-source models (via Ollama) for cost-flexible deployment.

Safe Security System | Python • YOLOv8 • OpenCV • face_recognition • Telegram API

- Trained a custom YOLOv8 model on a self-collected and manually annotated 80-image dataset using LabelImg.
- Built a danger zone system around the detected safe and tracked intruders with a 5-second timer.
- Integrated real-time face recognition (dlib) to distinguish authorized users from intruders.
- Automated alerts via Telegram Bot and Gmail SMTP with photo evidence on intrusion detection.

Customer Churn Prediction | Python • Scikit-learn • Pandas • Jupyter Notebook

- Applied and compared 4 ML algorithms: Random Forest, Decision Tree, KNN, Logistic Regression.
- Achieved 93.49% accuracy with a Voting Classifier ensemble on a 3,150-instance telecom dataset.
- Performed full ML pipeline: preprocessing, feature engineering, model evaluation (F1, RMSE, Accuracy).

Online Grocery Shop (Desktop App) | Python • Tkinter • SQLite3 • Pillow

- Designed a full-featured desktop application with Sign In, Sign Up, Shop, and Admin Panel.
- Implemented CRUD operations with SQLite3 for customer and warehouse management.
- Built admin panel with customer search, profile editing, and inventory control features.

EDUCATION

B.Sc. Computer Science | Islamic Azad University, Science and Research Branch

Currently in 6th Semester • 2022 – Present

CERTIFICATIONS

Computer Vision — Tehran Institute of Technology • Score: 96/100 • 2026

Machine Learning with Python — Tehran Institute of Technology • Score: 96/100 • 2026
Python Programming Advanced — Tehran Institute of Technology • Score: 95/100 • 2025

HIGHLIGHTS

Quick Learner ♦ Complex Problem Solver ♦ Innovative Thinker ♦ Strong Decision Maker ♦ Team Supporter

LANGUAGES

Persian — Native
English — B2/C1 (Upper Intermediate – Advanced)